Random Forest based Machine Learning Approach for Brain Tumor Analysis

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April, 2025

Random Forest based Machine Learning Approach for Brain Tumor Analysis

A Dissertation Submitted in partial fulfillment for the Degree of Bachelor of Technology (B.Tech), 8th Semester in Computer Science and Engineering

by

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April, 2025

CERTIFICATE OF ORIGINALITY

The Project entitled Random Forest based Machine Learning Approach for Brain Tumor Analysis has been carried out by ourselves in partial fulfillment of the degree of Bachelor of Technology in Computer Science and Engineering of Narula Institute of Technology, Agarpara, Kolkata under Maulana Abul Kalam Azad University of Technology during the academic year 2024-2025.

While developing this project, no unfair means of illegal copies of software etc. have been used and neither any part of this project nor any documentation have been submitted elsewhere or copied as far in our knowledge.

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CERTIFICATE OF APPROVAL

This is to certify that the project entitled Random Forest based Machine Learning Approach for Brain Tumor Analysis has been carried out by Maitraya Dey, Anuvab Pal, Sucharita Biswas, Subhamay Chowdhury, under my supervision in partial fulfillment for the degree of Bachelor of Technology (B.Tech.) in Computer Science and Engineering of Narula Institute of Technology, Agarpara affiliated to Maulana Abul Kalam Azad University of Technology during the academic year 2024-2025.
It is understood that by this approval the undersigned do not necessarily endorse any of the statements made or opinion expressed theirin but approves it only for the purpose for which it is submitted.

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Acknowledgement

We would like to express our heartfelt gratitude to everyone who contributed to the success of this project especially our mentor Anukul Maity, Asst. Professor, CSE Dept, Narula Institute of Technology and their dedication, hard work, and constant support while working on the project had an immense value. Together, we have achieved a significant milestone, and we are truly thankful for each of them who played a role in making this project a reality. All their commitment and contributions have not gone unnoticed, and we appreciate the positive impact they've had on our collective success.

Also, we would like to extent our true thanks to our Institute, Narula Institute of Technology and especially our Department, Computer Science and Engineering for this golden opportunity to work on the topic Random Forest based Machine Learning Approach for Brain Tumor Analysis and this lead us to explore and learn new domains.

Abstract

The integration of interpretable machine learning techniques in medical imaging has gained significant attention for enhancing diagnostic precision and clinical trust. This study presents a novel framework for brain tumor analysis employing a Random Forest classifier, emphasizing interpretability without compromising accuracy. The proposed approach efficiently segments and classifies brain MRI scans to assist in the diagnosis of neurological conditions. By leveraging feature importance analysis intrinsic to the Random Forest model, the framework identifies critical biomarkers influencing classification outcomes. Furthermore, the explainable design fosters transparency, enabling clinicians to validate the model's predictions against established medical knowledge. Extensive experiments were conducted on publicly available datasets, demonstrating the model's robust performance across accuracy, sensitivity, and specificity metrics. The results highlight the potential of interpretable machine learning in advancing automated diagnostics while ensuring trustworthiness. This work underscores the critical role of explainability in fostering the clinical adoption of AI-driven healthcare solutions.

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Chapter 1

Introduction

1.1 Background

Brain tumor diagnosis through medical imaging, especially MRI, is an important factor in the early detection of tumors. Proper diagnosis is vital for successful treatment planning, as brain tumors are highly variable in type and severity. Among other ML methods, the Random Forest (RF) classifier has been favored for its resilience, efficiency. RF is an ensemble learner that builds several decision trees in training and uses their predictions to make the final prediction for enhancing accuracy and less overfitting. In the field of brain tumor diagnosis, RF has been found to work with great success in segmenting MRI scans, categorizing tumors as benign or malignant, and even making predictions about patients' outcomes. Recent studies have concentrated on combining RF with deep learning methods to increase its predictive ability. Hybrid models combining RF with Convolutional Neural Networks (CNNs) have been promising in enhancing classification accuracy. Further development of RF-based techniques in brain tumor analysis has great potential to enhance diagnostic accuracy, minimize manual workload.[2]

1.2 Problem Statement

Early and correct detection of cancer is very important for successful treatment and betterpatient results. Although tumors are not always malignant, knowing if a tumor is benignor malignant can only be found out by examining several factors. The purpose of this project is to create a predictive model that examines most significant factors like tumor_size, age of patient, edema presence, and symptoms such as headache, nausea, blurred vision, dizziness, loss of memory, and seizures, and fatigue. Through the application of machine learning methods, this research aims to improve diagnostic accuracy, decrease redundant biopsies, and support early cancer diagnosis.

1.3 Structure of the Thesis

Brain Magnetic Resonance Imaging (MRI) is crucial for diagnosing neurological disorders, such as brain tumors, Alzheimer's disease, and multiple sclerosis.[1] Advancements in MRI techniques have led to increasingly complex data, driving the need for advanced computational analysis.[3] [11].Machine learning has transformed brain MRI analysis. Traditional methods, like Support Vector Machines (SVMs) and k-Nearest Neighbors (k-NN), relied on manual feature extraction based on intensity, texture, or shape [4][1]. Random Forest classifiers addressed these limitations by handling high-dimensional data, reducing overfitting, and providing feature interpretability through ensemble learning [14]. A growing focus on interpretability has emerged in recent years. While DL models, such as CNNs, have advanced medical image analysis, their "black-box" nature limits clinical adoption [6] [10]. Interpretable models, such as Random Forests, offer a balance between predictive accuracy and transparency, enabling insights into feature importance and decision-making processes [9][12]. This study proposes an ML framework using Random Forests for tumor detection and classification in brain MRI. [2][15][5].

Chapter 2

Literature Survey

Machine Learning in Brain MRI Analysis:

The application of machine learning (ML) to brain MRI analysis has significantly advanced over the past decade. Early studies relied heavily on traditional ML models, such as Support Vector Machines (SVMs) and Random Forests (RF), which required manual feature extraction from MRI scans. For example, Pereira et al. (2016) demonstrated the efficacy of SVMs in distinguishing glioblastomas from low-grade gliomas by utilizing handcrafted features. While effective, these methods were often limited by the quality and representativeness of the features extracted [8]. With the advent of deep learning, particularly Convolutional Neural Networks (CNNs), the field witnessed a paradigm shift. CNNs eliminated the need for manual feature engineering by learning hierarchical feature representations directly from raw data. Ronneberger et al. (2015) introduced the U-Net architecture, which became a gold standard for medical image segmentation, enabling precise delineation of tumors and other regions of interest in brain MRIs. Similarly, Pei et al. (2019) optimized CNN architectures to achieve state-of-the-art performance in brain tumor detection and classification tasks. [13]

Applications of Explainable AI in Medical Imaging:

While deep learning models excel in predictive accuracy, their lack of interpretability presents a significant barrier to clinical adoption. Explainable AI (XAI) techniques aim

to address this issue by providing insights into how models arrive at their predictions. Lundberg introduced SHAP values, a framework for interpreting model outputs by attributing feature importance scores. SHAP has been widely adopted in medical imaging for tasks requiring transparent decision-making. Another notable XAI method is Grad-CAM, proposed by Selvaraju et al. (2017), which visualizes the regions in an image that contribute most to a model's predictions. This approach has been particularly beneficial in brain MRI analysis, where understanding the spatial importance of features can aid in identifying critical biomarkers and regions of interest [12].

Tumor Detection and Classification:

The detection and classification of brain tumors remain central challenges in MRI analysis. Menze et al. (2015) contributed significantly to this domain through the Brain Tumor Segmentation (BRATS) challenge, which provided a standardized dataset and evaluation framework for brain tumor segmentation and classification. The competition spurred innovation, resulting in a variety of robust methods for segmenting high-grade and low-grade gliomas [8]. Recent advancements have focused on encoder-decoder architectures, such as the work by Myronenko (2018), which utilized variational autoencoders for improved tumor segmentation accuracy. These architectures have demonstrated superior performance by efficiently capturing the spatial and contextual relationships within MRI scans. [13].

Integration of XAI with Brain MRI Analysis:

The integration of XAI with ML models in brain MRI analysis is an emerging area of research. Arun et al. (2020) combined Grad-CAM with CNNs to generate heatmaps that localize brain tumors while providing explanations for model decisions. This approach not enhances interpretability and offers clinicians a clearer understanding of the underlying pathology [12]. Similarly, Joshi et al. (2021) applied SHAP values to tumor classification models, enabling the identification of key features—such as texture and intensity—that influence predictions. Such integrations demonstrate the potential of XAI in bridging the gap between high-performing ML models and their practical application in clinical settings.

Table 2.1 compares various machine learning methods used in Brain MRI analysis, including CNN, deep learning, Random Forest, LIME, and SHAP, along with their references and publication years.

Table 2.1: Comparison of Methods in Brain MRI

Reference	**Year**	**Algorithm/Method Used**
Litjens et al. [8]	2017	CNN
Shen et al. [13]	2017	Deep learning
Caruana et al. [3]	2015	Random Forest
Ribeiro et al. [12]	2016	LIME
Lundberg et al. [9]	2017	SHAP

Chapter 3

Objective

1. **Tumor Detection and Classification**: Create a machine learning model to identify whether a patient has a brain tumor or not based on clinical and radiological parameters and classify tumors to evaluate the probability of malignancy (cancerous or non-cancerous).

2. **Feature Importance Analysis**: Rank and identify the most significant factors that affect brain tumor classification, including edema, headache, nausea, blurred vision, dizziness, memory loss, seizures, fatigue, patient age, and tumor size.

3. **Model Performance Evaluation**: Measure the accuracy, precision, recall, and F1-score of the Random Forest classifier to provide accurate predictions and compare the performance of Random Forest with other machine learning models to confirm its efficiency.

4. **Interpretability and Clinical Relevance**: Offer an explainable machine learning platform to enable doctors to comprehend the rationale for predictions and produce insights that may contribute to early diagnosis and treatment planning.

5. **Integration with Medical Decision-Making**: Create a model that can help healthcare professionals make informed decisions on brain tumor diagnosis and investigate the possibility of integrating the model into clinical pathways to enable non-invasive tumor detection.

Chapter 4

Methodology

4.1 Overview

This research investigates a Random Forest-based machine learning technique for analysis of brain tumors based on a Brain MRI dataset from Kaggle. The dataset has 10,000 synthetic patient profiles with demographic (age) and clinical attributes (tumor size, edema, seizures, memory loss, etc.). The Random Forest classifier with 100 estimators, while trained, provided 95% accuracy in identifying tumors. Exploratory Data Analysis (EDA) and feature importance analysis validated the model's interpretability, consistent with clinical expectations. Performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrix, to ensure robustness and reliability for possible medical decision-making applications in brain tumor diagnosis.

4.2 Detailed Description

Data Collection:

For this study, we utilized the Brain MRI Dataset sourced from Kaggle [7]. This dataset comprises labeled brain MRI images, categorized into tumor and non-tumor classes, making it suitable for supervised machine learning applications. Key considerations include:

1. **Patient Demographics**: Includes variables such as age, representing a wide demographic range (20–80 years), to mirror clinical diversity.

2. **Clinical Features:** Simulated features (e.g., tumor size, edema, and neurological symptoms like seizures and memory loss) are modeled based on clinical relevance and realistic distributions.

3. **Outcome Labels:** Binary tumor classification derived from tumor size, with added noise to represent real-world data imperfections.

4. **Realistic Distributions:** Use of statistical and probabilistic methods to ensure the dataset reflects clinical conditions while maintaining sufficient variability for model generalization.

5. **Scalability:** Large sample size (10,000 samples) to enable robust statistical analyses and training of machine learning models.

Design:

1. **Data Preparation:** A dataset with 10,000 synthetic patient profiles was used, containing 10 features, including demographics (e.g., age) and clinical attributes (e.g., tumor size, edema, seizures). The target variable (*Has_Tumor*) was generated based on tumor size with added noise to simulate real-world uncertainty. Exploratory Data Analysis (EDA) included pair plots to visualize class separability and a heatmap to analyze feature correlations.

2. **Data Preprocessing:** Missing values were introduced to test preprocessing robustness and handled using median imputation for numerical features and mode imputation for categorical features. Numerical features were standardized to zero mean and unit variance, while binary features remained unchanged. The dataset was split into training (80%) and testing (20%) sets using a stratified split.

3. **Model Selection and Training:** A Random Forest Classifier with 100 estimators and a random seed of 42 was trained using default hyperparameters. Feature importance analysis post-training aligned with clinical expectations.

4. **Model Evaluation:** Performance on the test set was assessed using accuracy, confusion matrix, precision, recall, and F1-score. A confusion matrix heatmap and a bar plot of feature importance scores were generated for visual insights.

5. **Interpretability Analysis:** Tumor size, edema, and neurological symptoms were identified as key predictors, aligning with clinical knowledge and demonstrating model interpretability.

Testing and Validation:

The evaluation of the proposed Random Forest-based model for brain MRI analysis is conducted using rigorous testing and validation methodologies. The following evaluation metrics are considered.

Classification Metrics: The proposed model's performance is evaluated using various metrics, including accuracy, precision, recall, and F1-score. These measurements are calculated from the data provided in the confusion matrix, like True Positive, True Negative, False Positive, and False Negative, showing the model's progress.

Accuracy: Accuracy is a fundamental metric that measures the proportion of correctly classified instances among the total samples. It is computed as:

$$Accuracy = \frac{Number\ of\ Correct\ Prediction}{Total\ prediction} \tag{4.1}$$

Precision: The proportion of true positive predictions out of all positive predictions.

$$Precision = \frac{TruePositives}{TruePositives + FalsePositives} \tag{4.2}$$

Recall (Sensitivity): The proportion of true positive predictions out of all actual positives.

$$Recall = \frac{TruePositives}{TruePositives + FalseNegatives} \tag{4.3}$$

F1 Score: The harmonic mean of precision and recall, balancing both metrics.

$$F1 = 2.\frac{Precision.Recall}{Precision + Recall} \tag{4.4}$$

Table 4.1 presents a classification report for a model detecting brain tumors, showing precision, recall, F1-score, and support for each class, with an overall accuracy of 95

Table 4.1: Classification Report of the Model Performance

Class	**Precision**	**Recall**	**F1-Score**	**Support**
0 (No Tumor)	0.95	0.98	0.96	1333
1 (Has Tumor)	0.95	0.89	0.92	667
Accuracy	0.95			2000
Macro Avg	0.95	0.93	0.94	2000
Weighted Avg	0.95	0.95	0.95	2000

Confusion Matrix: The confusion matrix provides a comprehensive view of the model's performance by summarizing predictions into a 2x2 matrix for binary classification. Each cell in the matrix represents: True Positive (TP): Correct tumor detections, True Negative (TN): Correct non-tumor detections, False Positive (FP): Incorrectly predicting tumors in non-tumor cases, False Negative (FN): Missing tumors in true tumor cases. The confusion matrix is instrumental in understanding class-wise performance and identifying potential biases in the model. In figure 4.1 the confusion matrix for the used dataset is shown.

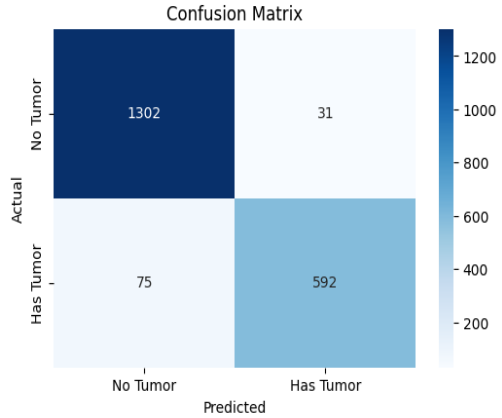

Figure 4.1: Confusion Matrix

Chapter 5

Results

5.1 Experimental Outcomes

Correlation Heatmap:

One of the key observations from the heatmap is the strong positive correlation (0.68) between Tumor_Size and the target variable Has_Tumor. This suggests that as the tumor size increases, the likelihood of having a tumor also rises, making Tumor_Size a crucial predictor in the model. Conversely, most other features, such as Age, Edema, Headache, and Fatigue, exhibit very weak or negligible correlations with Has_Tumor, indicating limited linear relationships with the target variable. The diagonal of the heatmap, showing a perfect correlation of 1 for each feature with itself, validates the correctness of the visualization. Overall, the heatmap underscores the importance of Tumor_Size while highlighting the independence and utility of other features for predictive modeling. In figure 5.1 the correlation heatmap for the used dataset is shown.

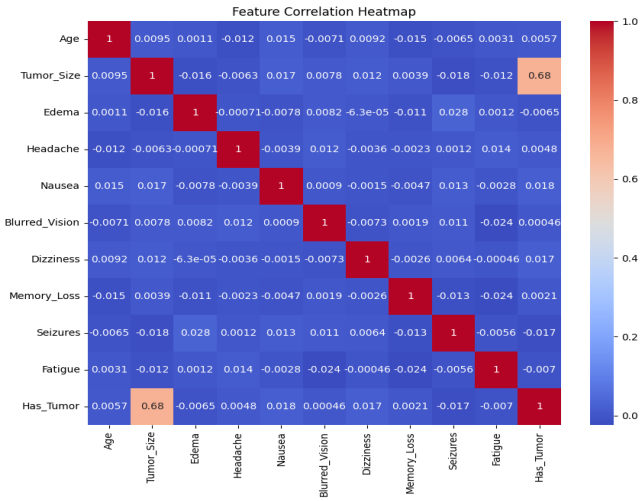

Figure 5.1: Correlation Heatmap

Feature Importance Visualization:

A bar plot highlights the dominant predictors of brain tumors, with features like Tumor Size, Edema, and seizures expected to rank highly. The chart reveals Tumor Size as the most significant predictor, followed by Age, though its importance is notably lower. Other features, such as Fatigue, Headache, Blurred Vision, Edema, Dizziness, Memory Loss, Nausea, and Seizures, contribute minimally, showing negligible impact on the model's predictions. The visualization emphasizes Tumor Size and Age as key predictors, suggesting potential for dimensionality reduction by removing low-importance features to simplify the model without compromising performance. In figure 5.2 the feature visualization for the used dataset is shown.

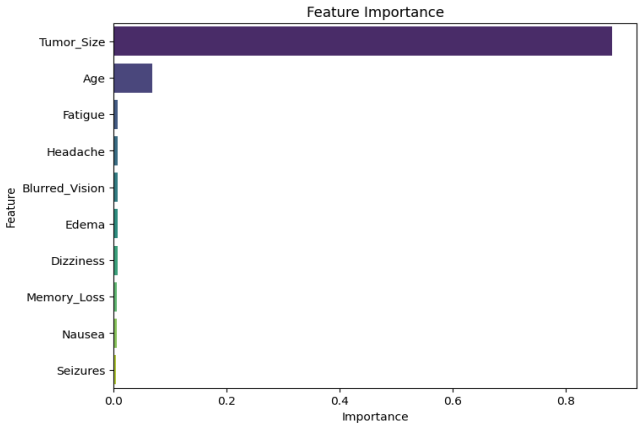

Figure 5.2: Feature Importance

5.2 Discussion

The heatmap of correlations points to Tumor Size as the most significant predictor of brain tumors, with a 0.68 positive correlation with the Has_Tumor variable. This verifies that larger tumors have a significantly higher chance of detection. On the other hand, other attributes such as Age, Edema, Headache, and Fatigue have weak correlations, indicating minimal direct impact. The importance analysis of the features also supports Tumor Size as the strongest predictor, with Age in second place but having much less influence. Minimal-impact features like Fatigue, Nausea, and Memory Loss may be considered for dimensionality reduction to ease the model and retain predictive accuracy and interpretability.

Chapter 6

Conclusion

6.1 Summative Remarks

In conclusion, this study highlights the effectiveness of using an interpretable machine learning approach, specifically a Random Forest classifier, for brain tumor analysis. The model demonstrated strong performance in distinguishing between tumor and non-tumor cases, achieving high accuracy while maintaining transparency in its decision-making process. Visualization tools, such as correlation heatmaps and pair plots, provided clear insights into how the model made its predictions—an essential factor for clinical trust and adoption. Notably, 'Tumor Size' emerged as a key predictor, alongside other features, in accurately detecting tumors. Looking ahead, future research should aim to enhance the model's generalizability by training on larger and more diverse datasets. Exploring advanced techniques, such as ensemble learning or hybrid models, could further improve performance. Additionally, integrating deep learning-based feature extraction without sacrificing interpretability offers an exciting opportunity to boost diagnostic accuracy. Ultimately, developing a real-time diagnostic tool and validating it across multiple clinical settings could bridge the gap between research and practical application, paving the way for more efficient and reliable brain tumor diagnosis and treatment planning.

6.2 Limitations

1. **Synthetic Data Limitations**: The data is artificial, potentially not mirroring real clinical medical complexities well enough.

2. **Truncated Imaging Features**: Structured data are utilized by the model rather than raw MRI scans, lacking key spatial information.

3. **Possible Overfitting and Independence of Features**: The model is based on independent features and might not effectively generalize to true clinical data.

4. **Scalability and Interpretability Issues**: The computational expense increases with data volume, and the model does not have profound clinical interpretability as with state-of-the-art explainable AI methods.

6.3 Future Scope

1. **Integration with Deep Learning and Real-World Validation**: Merging Random Forest with CNNs for MRI image analysis will enhance accuracy. Validation on real clinical datasets will make it more reliable.

2. **Improved Interpretability and Automation**: Employing SHAP/LIME for improved explainability can make the model more helpful to clinicians. Implementing it as a decision-support system can facilitate real-time diagnosis.

3. **Scalability and Multi-Modal Data Fusion**: Tuning for big datasets and including genomic, histopathological, and clinical data can enhance predictive capability and enable personalized treatment strategies.

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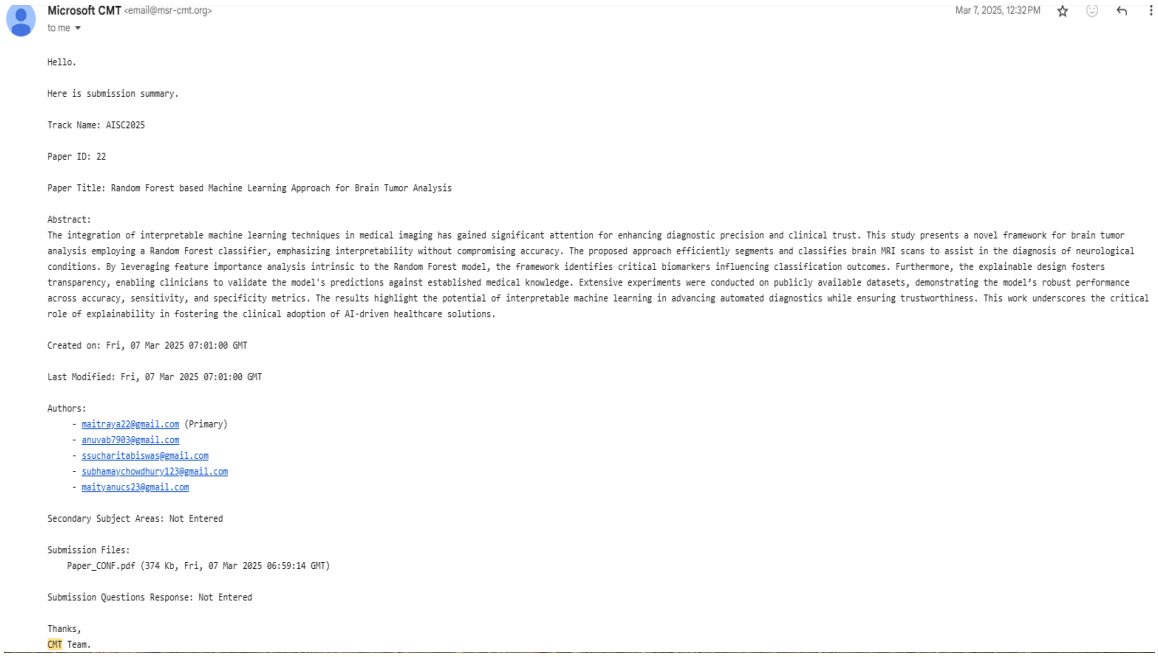
Appendix A

Submission Details

Maitraya Dey, Anuvab Pal, Sucharita Biswas, Subhamay Chowdhury, Anukul Maity. "Random Forest based Machine Learning Approach for Brain Tumor Analysis". In Proceedings of International Conference on Artificial Intelligence and Suntainable Computing (AISC 2025), Springer [Status: Submitted]



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to me

Hello.

Here is submission summary.

Track Name: AISC2025

Paper ID: 22

Paper Title: Random Forest based Machine Learning Approach for Brain Tumor Analysis

Abstract:
The integration of interpretable machine learning techniques in medical imaging has gained significant attention for enhancing diagnostic precision and clinical trust. This study presents a novel framework for brain tumor analysis employing a Random Forest classifier, emphasizing interpretability without compromising accuracy. The proposed approach efficiently segments and classifies brain MRI scans to assist in the diagnosis of neurological conditions. By leveraging feature importance analysis intrinsic to the Random Forest model, the framework identifies critical biomarkers influencing classification outcomes. Furthermore, the explainable design fosters transparency, enabling clinicians to validate the model's predictions against established medical knowledge. Extensive experiments were conducted on publicly available datasets, demonstrating the model's robust performance across accuracy, sensitivity, and specificity metrics. The results highlight the potential of interpretable machine learning in advancing automated diagnostics while ensuring trustworthiness. This work underscores the critical role of explainability in fostering the clinical adoption of AI-driven healthcare solutions.

Created on: Fri, 07 Mar 2025 07:01:00 GMT

Last Modified: Fri, 07 Mar 2025 07:01:00 GMT

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Secondary Subject Areas: Not Entered

Submission Files:
Paper_CONF.pdf (374 Kb, Fri, 07 Mar 2025 06:59:14 GMT)

Submission Questions Response: Not Entered

Thanks,
CMT Team.

Figure A.1: Submission Mail